

- 6 (a)** Starting from the definitions of  $\tanh$  and  $\operatorname{sech}$  in terms of exponentials, prove that

$$1 - \tanh^2 u = \operatorname{sech}^2 u. \quad [3]$$

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

- (b)** Show that  $\frac{d}{dt}(\operatorname{sech}^{-1} t) = -\frac{1}{t\sqrt{1-t^2}}$ . [4]

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It is given that

$$x = \tanh^{-1}t \quad \text{and} \quad y = t \operatorname{sech}^{-1}t, \quad \text{for } 0 < t < 1.$$

- (c) Show that  $\frac{dy}{dx} = -\sqrt{1-t^2} + (1-t^2)\operatorname{sech}^{-1}t$ . [4]

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- (d) Find  $\frac{d^2y}{dx^2}$  in terms of  $t$ . [4]

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